CHAPTER

7

Chemical Equilibrium + Acids Bases & Salts

For NSEJS 2025

INTRODUCTION & TYPES OF EQUILIBRIUM:

(a) Reversible Reactions:

The chemical reactions which can proceed both in the forward and reverse directions under similar conditions are called reversible reactions.

(i) They are indicated by a double half headed arrow ($\rightleftharpoons$).

(ii) Change in temperature, pressure or concentration changes the direction of reaction.

(iii) Such reactions are carried out in a closed container.

(iv) Such reactions never go to completion.

e.g. $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$ (thermal dissociation)

$C_2H_5OH_{(\ell)} + CH_3COOH_{(\ell)} \rightleftharpoons CH_3COOC_2H_{5(aq)} + H_2O_{(\ell)}$ (esterification)

$FeCl_{3(s)} + 3H_2O_{(\ell)} \rightleftharpoons Fe(OH)_{3(aq)} + 3HCl_{(aq)}$ (hydrolysis)

(b) Irreversible Reactions:

The reactions in which complete conversion of the reactants into products occur are called irreversible reactions.

(i) They are indicated by a single headed arrow. ($\rightarrow$)

(ii) Change in concentration, temperature and pressure doesn't change the direction of reaction.

(iii) They can proceed only in the forward direction.

e.g. $2KClO_{3(s)} \xrightarrow{Heat} 2KCl_{(s)} + 3O_{2(g)}$

(Thermal decomposition)

$NaOH_{(aq)} + HCl_{(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{(\ell)}$

(neutralization)

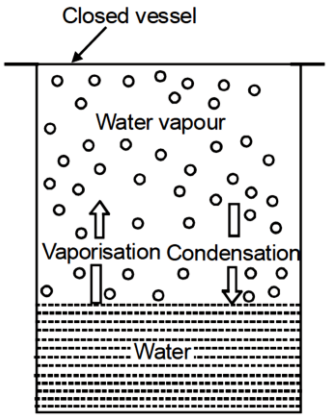
$BaCl_{2(aq)} + H_2SO_{4\,(dil)} \longrightarrow BaSO_{4(s)} + 2HCl_{(aq)}$

(precipitation)

$2FeCl_3 + SnCl_2 \longrightarrow 2FeCl_2 + SnCl_4$ (redox)

(c) Physical equilibrium

Equilibrium is also attained in physical processes such as vaporization and condensation. The conversion of a liquid into vapour is known as vaporization, while the opposite process, i.e. the conversion of a vapour into liquid is called condensation. When the vessel containing a liquid is tightly closed, the liquid does not evaporate completely. This is because the molecules constituting the vapour can not go out of the vessel. Most of them condense and return to the liquid. A state is soon reached when the number of molecules leaving the surface of the liquid equals the number of molecules returning to the liquid phase in per unit time. At this equilibrium state, the number of molecules present in the vapour phase and liquid phase become fixed.



(d) Chemical equilibrium

A chemical equilibrium may be defined as a state of a reversible reaction when the observable properties such as colour, concentration etc., do not show any noticeable change.

or The state of a reversible reaction when the rates of both forward & backward reaction become equal.

(i) Dynamic Equilibrium:

All the observable properties of the system become constant at dynamic equilibrium. It may lead us to think that the reaction stops altogether at equilibrium and that equilibrium is static in nature. But this is not true. Actually, at equilibrium, the rate of forward reaction becomes equal to the rate of backward reaction so that there is no net change in the concentrations of various species. In other words, we can say that the equilibrium state is a dynamic balance between the forward and the backward reaction.

This can be illustrated by considering the reaction between hydrogen and iodine to form hydrogen iodide.

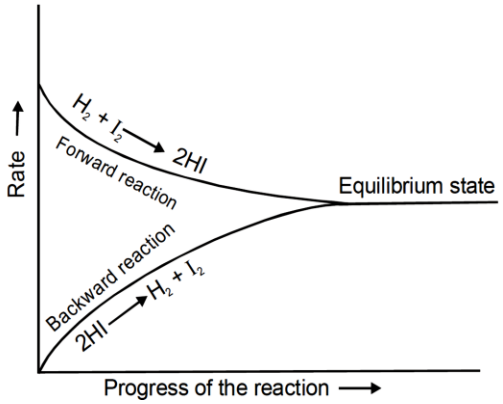
When hydrogen and iodine are taken in a closed vessel maintained at 444ºC, hydrogen molecules combine with iodine molecules to form hydrogen iodide.

$H_2(g)$	+	$I_2(g)$	$\longrightarrow$	2HI(g)
Hydrogen		Iodine		Hydrogen Iodide

Since the reaction is reversible in nature, the molecules of HI formed also begin to dissociate to form H_2 and I_2.

$2HI(g) \longrightarrow H_2(g) + I_2(g)$

The variation of rates of forward and backward reaction have been shown in the figure. After the attainment of equilibrium, change in concentrations occurs provided the temperature of the reaction mixture is kept constant. Thus, at equilibrium, the reaction does not stop but the system acquires constant observable properties because of the equal rates of forward and backward reactions. Thus, the equilibrium is dynamic in nature.



Attainment of equilibrium state

Characteristics of Dynamic Equilibrium:

(i) At equilibrium, rate of forward reaction is equal to that of backward reaction.

(ii) At equilibrium observable properties (Macroscopic) such as concentration, density, colour etc. are constant under the given conditions of temperature and pressure.

(iii) At equilibrium, microscopic properties involving processes continue but they are in balance.

(iv) Equilibrium can be attained from either direction of reaction beginning with reactants or products.

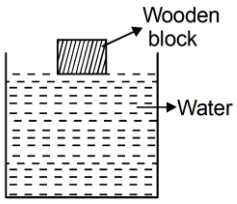
(v) Equilibrium can be attained in a closed system only.

(vi) A catalyst leads to early attainment of a chemical equilibrium. As it accelerates both the forward and the backward reactions to the same extent, thus it does not disturb the equilibrium point.

(ii) Static Equilibrium:

A system is in static equilibrium when various forces acting on it produce no resultant force and therefore, system remains motionless.

e .g. Wooden block floating on water



STATES OF CHEMICAL EQUILIBRIUM

State of equilibrium means the balance of driving forces i.e. the factors taking the reaction in forward direction and the backward direction are balancing each other.

The equilibrium state represents a compromise between two opposing tendencies.

- Molecules try to minimize energy.
- Molecules try to maximize entropy.

In a reversible reaction like–

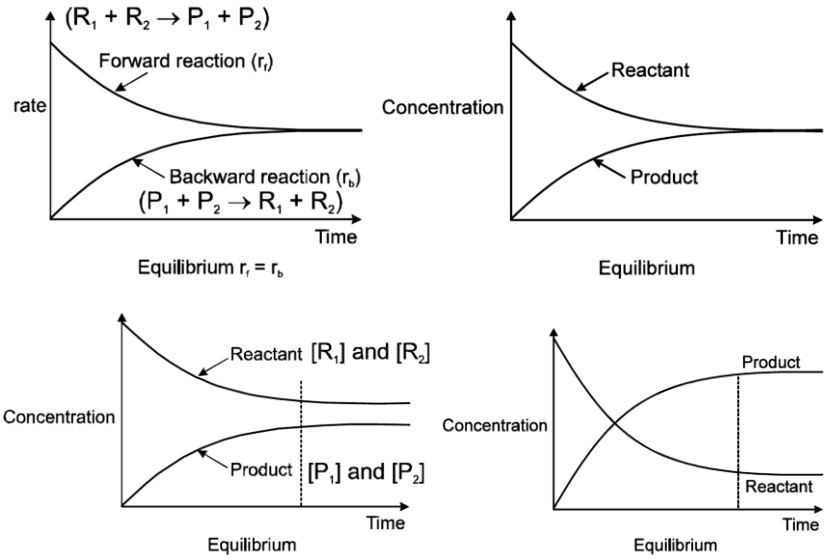
Initially only reactants are present. R_1 and R_2 combine to form P_1 and P_2. As soon as P_1 and P_2 are formed**,** they start the backward reaction. As concentrations of R_1 and R_2 decrease rate of forward reaction decreases and rate of backward reaction increases. Ultimately a stage is reached when both the rates become equal. Such a state is known as "Chemical Equilibrium" or "state of Equilibrium".

At equilibrium:

(i) Rate of forward reaction (r_f) = rate of backward reaction (r_b)

(ii) Concentration (mole/litre) of reactant and product remains constant with respect to time.

$$\underbrace{R_1 + R_2}_{\text{Reactants}} \rightleftharpoons \underbrace{P_1 + P_2}_{\text{Products}}$$



Law of mass action

The law was introduced by Guldberg & Waage in 1864.

According to the law of mass action at a given temperature, the rate of a chemical reaction is proportional to the product of the active masses of the reactants.

Consider a reaction -

$$A + B \rightleftharpoons C + D$$

Rate of forward reaction α [A][B]

$r_f = K_f [A][B]$ {K_f = Rate constant for forward reaction}

Similarly,

$r_b = K_b [C][D]$ {K_b = Rate constant for backward reaction}

At equilibrium $r_f = r_b$

So, $K_f [A][B] = K_b [C][D]$

$$\frac{K_f}{K_b} = \frac{[C][D]}{[A][B]}$$

$$\therefore K_C = \frac{[C][D]}{[A][B]} \left(K_C = \frac{K_f}{K_b} \right)$$

Here K_C is called the equilibrium constant of the reaction.

Therefore, at constant temperature, the ratio between the rate constant of forward reaction and the rate constant of backward reaction is called equilibrium constant.

For a general reaction:

$aA + bB \rightleftharpoons cC + dD$

K_C may be written as:

$$K_C = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

If reactants & products are gaseous then,

$$K_P = \frac{(P_C)^c \times (P_D)^d}{(P_A)^a \times (P_B)^b}$$

Where P_A, P_B, P_C, P_D are partial pressures of gas A, B, C, D respectively.

$$\text{Partial Pressure} = \frac{\text{Number of moles of reactant}}{\text{Total moles in the reaction mixture}} \times \text{Total Pressure}$$

Relation between K_p & K_c:

$K_p = K_c (RT)^{\Delta n}$

(a) If $\Delta n = 0$, then $K_p = K_c$

(b) Δn = + ve, then $K_p > K_c$

(c) Δn = – ve, then $K_p < K_c$

❖ **Mind it:** Rate constant of a chemical reaction is equal to the rate of the chemical reaction while the concentration of reactants are unity.

Significance of n

$$n = \frac{\text{sum of sotichiometric coefficient of product}}{\text{sum of cofficient of reactants}}$$

(i) for $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ (n = 2)

(ii) for $2NH_{3(g)} \rightleftharpoons N_{2(g)} + 3H_{2(g)}$ $\left(n = \frac{3}{2} + \frac{1}{2} = 2\right)$

(iii) for $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$ (n = 1)

Example: Find out K_C of $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

Solution: $K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$

Example: Find out K_C of $N_2 + 3H_2 \rightleftharpoons 2NH_3$

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

Example: For the reaction, $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(A) $K_p = -K_c$ (B) $K_c = 0$ (C) $K_p = K_c$ (D) $K_p = 0$

Solution: For the reaction having $\Delta n = 0$, $K_p = K_c$

Example: For the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, Find out Δn

Solution: Δn = moles of product – moles of reactant

$\Delta n = 2-(3 + 1)$

$\Delta n = -2$

EQUILIBRIUM CONSTANT

(i) The equilibrium constant (K_c) is a constant for any given chemical reaction at equilibrium at a definite temperature.

(ii) The value of K_C is different for different reaction at equilibrium. It is also different for the same reaction carried out at different temperatures.

(iii) The change in either pressure or concentration at equilibrium may change the state of equilibrium but it cannot change the equilibrium constant which depends only on temperature.

(iv) The value of K_C indicates the extent to which a chemical reaction can occur.

Larger the value of K_C for a particular reaction, higher will be the concentration (or amount) of the product formed. On the other hand, a small value of K_C indicates that the concentration of the product at equilibrium will be low.

e.g. $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

The value of K_C for the above reaction is 1.9 at 298 K.

This low value indicates that at equilibrium the concentration of PCl_5 will be low.

Similarly, the value of K_C for the reaction -

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, is 3.6×10^8 at 298 K,

The large value of K_C indicates that a major portion of N_2 and H_2 will be combined to form NH_3.

(a) Effect of Mode of Representation on Equilibrium constant

Let us consider the following reaction at equilibrium

$2SO_2 + O_2 \rightleftharpoons 2SO_3$

The equilibrium constant for the reaction (say K_1) is given by -

$$K_1 = \frac{[SO_3]^2}{[SO_2]^2[O_2]} \qquad \ldots(i)$$

If we reverse the representation of above reaction

$2SO_3 \rightleftharpoons 2SO_2 + O_2$

then the equilibrium constant for the reaction (say $\mathbf{K_2}$) is given by -

$$K_2 = \frac{[SO_2]^2[O_2]}{[SO_3]^2}$$

$$\text{or } \frac{1}{K_2} = \frac{[SO_3]^2}{[SO_2]^2[O_2]} \qquad \ldots(ii)$$

Comparing (i) and (ii) we get $K_1 = \frac{1}{K_2}$

❖ **Mind it:** Thus, the equilibrium constants of the reactions are reciprocal of each other.

(b) Application of Equilibrium Constant

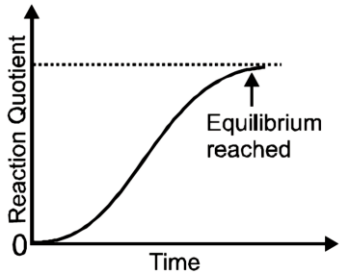
- **Predicting the direction of the reaction**

Reaction Quotient (Q)

At each point in a reaction, we can write a ratio of concentration terms having the same form as the equilibrium constant expression. This ratio is called the reaction quotient denoted by symbol Q.

It helps in predicting the direction of a reaction.

The expression $Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$ at any time during reaction is called reaction quotient. The concentrations [C], [D], [A], [B] are not necessarily at equilibrium.



- The reaction quotient is a variable quantity with time.
- It helps in predicting the direction of a reaction.

◈ if Qc > Kc reaction will proceed in backward direction until equilibrium in reached.

◈ if Qc < Kc reaction will proceed in forward direction until equilibrium is established.

◈ if Qc = Kc Reaction is at equilibrium.

eg. $2A(g) + B(g) \rightleftharpoons C(g) + D(g)$

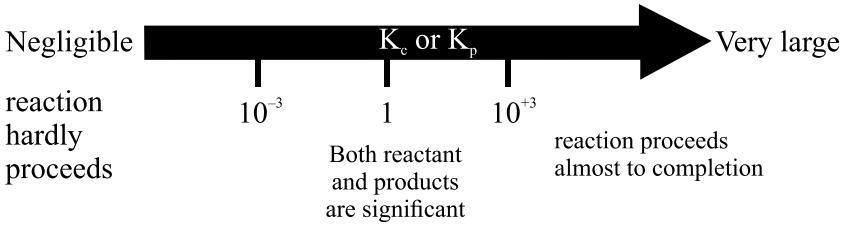
Q_C = Reaction quotient in terms of concentration

$$Q_C = \frac{[C][D]}{[A]^2[B]}$$

$$K_C = \frac{[C]_{eq}[D]_{eq}}{[A]^2_{eq}[B]_{eq}}$$ [Here all the conc. are at equilibrium]

- **Predicting the extent of the reaction**

$$K = \frac{[\text{Product}]}{[\text{Reactant}]}$$



Case-I If K is large ($k > 10^3$) then product concentration is very-very larger than the reactant ([Product] >>[Reactant]) Hence concentration of reactant can be neglected with respect to the product. In this case, the reaction is product favorable and equilibrium will be more in forward direction than in backward direction.

Case-II If K is very small ($K < 10^{-3}$)

[Product] << [Reactant]

Hence concentration of Product can be neglected as compared to the reactant.

In this case, the reaction is reactant favorable.

(c) Characteristic of Equilibrium Constant & Factors Affecting It

- Equilibrium constant does not depend upon concentration of various reactants, presence of catalyst, direction from which equilibrium is reached
- The equilibrium constant does not give any idea about time taken to attain equilibrium.
- **K depends on the stoichiometry of the reaction.**

◈ If two chemical reactions at equilibrium having equilibrium constants K_1 and K_2 are added then the resulting equation has equilibrium constant $K = K_1 . K_2$

Equation constant

$A(g) \rightleftharpoons B(g) \quad K_1$

$B(g) \rightleftharpoons C(g) \quad K_2$

On adding $A(g) \rightleftharpoons C(g) \qquad K = K_1 . K_2$

◈ If the reaction having eq. constant K_1 is reversed then resulting equation has eq. constant $\frac{1}{K_1}$

$A(g)+B(g) \rightleftharpoons C(g) + D(g)$

On reversing, $C(g)+D(g) \rightleftharpoons A(g) + B(g)$ $K = \frac{1}{K_1}$

◈ If a chemical reaction having equilibrium constant K_1 is multiplied by a factor n then the resulting equation has equilibrium constant $K = (K_1)^n$, n can be fraction

eg. $D_2(g) \rightleftharpoons 2A(g)$ K_1 Multiplying by $\left(\frac{1}{2}\right)$

$\frac{1}{2} D_2(g) \rightleftharpoons A(g)$ $K = (K_1)^{1/2} = \sqrt{K_1}$

- **Equilibrium constant is dependent only on the temperature.**

It means k_p and k_c will remain constant at constant temperature no matter how much changes are made in pressure, concentration, volume or catalyst.

◈ However if temperature is changed,

$\text{Log}\ \frac{K_2}{K_1} = \frac{\Delta H}{2.303 R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$ (Vant Hoff Equation); ΔH = Enthalpy of reaction

If $T_2 > T_1$ then

$K_2 > K_1$ provided ΔH = +ve (endothermic reaction)

$K_2 < K_1$ if ΔH = –ve (exothermic reaction)

In the above equation, the unit of R and $\Delta H/T$ should be same.

- **Relation between equilibrium constant & standard free energy change.**

  $\Delta G^o = -2.303\ RT \log K$

  Where $\Delta G°$ = standard free energy change
  T = Absolute temperature,
  R = universal gas constant.

(d) Homogeneous liquid system: Formation of ethyl acetate:

"Homogeneous equilibria, in which all reactants & products are in a single phase, usually either gaseous phase or solution."

The reaction between alcohol and acid to form ester is an example of homogeneous equilibrium in liquid system.

$CH_3COOH(\ell) + C_2H_5OH(\ell) \rightleftharpoons CH_3COOC_2H_5(\ell) + H_2O(\ell)$

$$K_C = \frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]}$$

Example: In an experiment starting with 1 mole of ethyl alcohol, 1 mole of acetic acid and 1 mole of water at T^0C, the equilibrium mixture on analysis shows that 54.3% of the acid is esterified. Calculate the equilibrium constant of this reaction.

Solution: $CH_3COOH(\ell) + C_2H_5OH(\ell) \rightleftharpoons CH_3COOC_2H_5(\ell) + H_2O(\ell)$

Initial	1	1	0	1
At equilibrium	1 – x	1 – x	x	1 + x
	1–0.543	1– 0.543	0.543	1+ 0.543

(54.3% of 1 mole = $\frac{1\times 54.3}{100}$ = 0.543 mole)

Hence given x = 0.543 mole

Applying law of mass action:

$$K_C = \frac{[\text{ester}][\text{water}]}{[\text{acid}][\text{alcohol}]} = \frac{0.543\times 1.543}{0.457\times 0.457} = 4.0$$

◈ EQUATION INVOLVING IONS:

Equilibrium involving ions always take place in aqueous medium. In case of expression of K_C concentration of ion is taken.

Ex. $Ag^+(aq.) + Cl^-(aq.) \rightleftharpoons AgCl(s)$ $\qquad K_c = \frac{1}{[Ag^+][Cl^-]}$

(e) Heterogenous Equilibrium:

"Hoterogeneous equilibria, in which all reactants & products are present in more than one phase."

For pure solid and pure liquid, active mass is taken to be unity i.e. 1 as they remain constant throughout the reaction

- $CaCO_{3\,(s)} \rightleftharpoons CaO_{(s)} + CO_{2\,(g)}$

  $K_P = P_{CO_2}$, $K_C = [CO_{2\,(g)}]$

$$[CaCO_{3(s)}] = \frac{\text{moles}}{\text{volume}} = \frac{\frac{W_{CaCO_3}}{M_{CaCO_3}}}{V} = \frac{\text{density } CaCO_3}{M_{CaCO_3}} = \text{constant}$$

$$K = \frac{[CaO(s)][CO_2(g)]}{[CaCO_3(s)]}$$

$$\frac{K.[CaCO_3(s)]}{[CaO(s)]} = [CO_2(g)]$$

$K_C = [CO_{2(g)}]$

- $H_2O_{(1)} \rightleftharpoons H_2O_{(g)}$

  $K_P = P_{H_2O(g)}$, $K_C = [H_2O_{(g)}]$

  [For pure solid and pure liquid active mass is taken as unity i.e. = 1]

Example: In a reaction $C_{(s)} + CO_{2\,(g)} \rightleftharpoons 2CO_{(g)}$, the equilibrium pressure is 12 atm. If 50% of CO_2 reacts. Calculate K_P.

Solution:

	$C_{(s)}$ +	$CO_{2\,(g)}$ $\rightleftharpoons$	$2CO_{(g)}$
t = 0		a	0
t = teq		$a - \frac{a}{2}$	$2\frac{a}{2}$

P_{eq} = 12 atm

$X_{CO_2} = \frac{1}{3}$, $X_{CO} = \frac{2}{3}$

$P_{CO_2} = \frac{1}{3} \times 12 = 4$

$P_{CO} = \frac{2}{3} \times 12 = 8$

$\therefore K_P = \frac{8 \times 8}{4} = 16$

DEGREE OF DISSOCIATION

It is the fraction of one mole dissociated into the products.

(Defined for one mole of substance)

So, α = no. of moles dissociated / initial no. of moles taken = fraction of moles dissociated out of 1 mole.

Mind it: % dissociation = α x 100

Suppose 5 moles of PCl_5 is taken and if 2 moles of PCl_5 dissociated then $\alpha = \frac{2}{5} = 0.4$

Let a gas An dissociates to give n moles of A as follows-

$A_{n\,(g)} \rightleftharpoons n\,A_{(g)}$

t = 0	a	0
$t = t_{eq}$	a – x	n.x

$\alpha = \frac{x}{a} \Rightarrow \quad x = a\alpha.$

$a - a\,\alpha = a(1-\alpha) \qquad n\,a\,\alpha$

Total no. of moles = $a - a\,\alpha + n\,a\,\alpha = [1 + (n-1)\,\alpha]\,a$

Example: Calculate the degree of dissociation and Kp for the following reaction.

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

t = 0	a	0	0
t = t	a –x	x	x

Since for a mole, x moles are dissociated

Solution: $\therefore$ For 1 mole, $\frac{x}{a}$ moles = α are dissociated

$\therefore x = a\,\alpha$

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

$t = t_{eq}$	$a - a\alpha$	$a\alpha$	$a\alpha$

Total no. of moles at equilibrium = $a + a\alpha = a\,(1 + \alpha)$

$$P_{PCl_5} = \frac{a(1-\alpha)P}{a(1+\alpha)}, \quad P_{PCl_3} = \frac{a\,\alpha.P}{a(1+\alpha)},$$

$$P_{Cl_2} = \frac{a\,\alpha}{a(1+\alpha)}.P$$

$$K_P = \frac{\left\{\left(\frac{\alpha P}{1+\alpha}\right)\right\}^2}{\left(\frac{1-\alpha}{1+\alpha}\right)P} \qquad K_P = \frac{\alpha^2.P}{1-\alpha^2}$$

EXTERNAL FACTORS AFFECTING EQUILIBRIUM

Le Chatelier's Principle:

If a change is applied to the system at equilibrium, then equilibrium will be shifted in that direction in which it can minimize the effect of change applied and the equilibrium is established again under new conditions.

- **Effect of concentration:** If the concentration of a component is increased, reaction shifts in a direction which tends to decrease its concentration. e.g. In the following example.

  $N_{2\,(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

  [reactant] ↑ Forward shift

  [Product] ↑ Backward shift

- ◈ If concentration of reactant is increased at equilibrium then reaction shifts in the forward direction.
- ◈ If concentration of product is increased then reaction shifts in the backward direction

Mind it: The addition of any solid component does not affect the equilibrium.

- **Effect of volume:**

◈ If volume is increased, pressure decreases hence reaction will shift in the direction in which pressure increases that is in the direction in which number of moles of gases increases and vice versa.

◈ If volume is increased then, for $\Delta n_g > 0$ reaction will shift in the forward direction

$\Delta n_g < 0$ reaction will shift in the backward direction

$\Delta n_g = 0$ reaction will not shift. eg. $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ (No effect)

Explanation:

(i) $\Delta n_g > 0$, eg. $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2\,(g)}$

$$Q_C = \frac{\frac{(n_{Cl_2})}{V} \times \frac{(n_{PCl_3})}{V}}{\frac{(n_{PCl_5})}{V}} \Rightarrow Q_C \propto \frac{1}{V} \text{ for } \Delta n_g > 0$$

On increasing V, Q_C, decreases.

Now, for $Q_C < K_C$ reaction will shift in forward direction.

Thus, if, Volume $\uparrow$ $Q_C \downarrow$ (Forward shift)

Volume $\downarrow$ $Q_C \uparrow$ (Backward shift)

(ii) $\Delta n_g < 0$, eg. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

$$Q_C = \frac{\left\{\frac{(n_{NH_3})}{V}\right\}^2}{\left\{\frac{(n_{N_2})}{V}\right\}\left\{\frac{(n_{H_2})}{V}\right\}^3}$$

$\Rightarrow Q_C \propto V^2$ for $\Delta n_g < 0$

$V \uparrow$ $Q_C \uparrow$ (Backward shift) ;

$V \downarrow$ $Q_C \downarrow$ (Forward shift)

- **Effect of pressure:**

On increasing pressure, equilibrium will shift in the direction in which pressure decreases i.e. no. of moles in the reaction decreases and vice versa.

$\because$ $P \propto$ no. of moles

(i) For $\Delta n_g = 0 \rightarrow$ No. effects

(ii) For $\Delta n_g > 0$, $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

$$Q_p = \frac{(X_{PCl_3}P).(X_{Cl_2}P)}{(X_{PCl_5}.P)} \Rightarrow Q_P \propto P \text{ [X = mole fraction]}$$

$P \downarrow$; $Q_P \downarrow$; (Forward shift) [P = Total pressure at equilibrium]

$P \uparrow$; $Q_P \uparrow$; (Backward shift)

(iii) For $\Delta n < 0$, eg. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$Q_P = \frac{\left[(X_{NH_3})P\right]^2}{\left[(X_{N_2}).P\right]\left[(X_{H_2})P\right]^3} \Rightarrow Q_P \propto \frac{1}{P^2}$$

$P \uparrow$; $Q_P \downarrow$; (Forward shift) ; $P \downarrow$; $Q_P \uparrow$; (Backward shift)

- **Effect of catalyst:**

  Due to catalyst, the state of equilibrium is not affected i.e. no shift will occur as catalyst lowers the activation energy of both the forward & reverse reaction by same amount, thus altering the forward & reverse rate equally and hence, the equilibrium will be attained faster i.e time taken to reach the equilibrium is less.

- **Effect of inert gas addition:**

(i) At constant volume: Inert gas addition has no effect at constant volume

(ii) At constant pressure: If inert gas is added then to maintain the pressure constant, volume is increased. Hence equilibrium will shift in the direction in which larger no. of moles of gas is formed

(i) $\Delta n_g > 0$, reaction will shift in the forward direction

(ii) $\Delta n_g < 0$, reaction will shift in the backward direction

(iii) $\Delta n_g = 0$, no effect

- **Effect of temperature:**

(i) Exothermic reaction: The reaction in which heat is evolved

$A(g) + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)} + \text{Heat}\ \Delta H = -\text{ve}$

eg. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} + \text{Heat}$

$T \uparrow \Rightarrow K'$ will decrease (from vant' hoff equation)

$$\log \frac{K_1}{K_2} = \frac{\Delta H^o}{2.303R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\log \frac{K_1}{K_2} < 0 \Rightarrow \log K_1 - \log K_2 > 0 \Rightarrow \quad \log K_1 > \log K_2 \Rightarrow K_1 > K_2$$

Reaction will shift in backward direction.

$T \downarrow \Rightarrow K$ will increases.

Reaction will shift in forward direction.

(ii) Endothermic reaction: energy consumed.

$A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)} - \text{Heat}$

$\therefore \Delta H = +\text{ve}$

$T \uparrow \Rightarrow K \uparrow \Rightarrow$ Forward ; $T \uparrow$

$\Rightarrow K \downarrow \Rightarrow$ Backward

Example: Which among of the following reaction will be favoured at low pressure?

(A) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

(B) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(C) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(D) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Solution: (C) On lowering the pressure, equilibrium favours the direction of higher volume.

Example: Which among the following reaction is favoured in forward direction by increase of temperature?

(A) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH3(g) + 22.9$ kcal

(B) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g) - 42.8$ kcal

(C) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) + 45.3$ kcal

(D) $H_2(g) + Cl_2(g) - 44$ kcal $\rightleftharpoons 2HCl(g)$

Solution: (B) This is endothermic reaction, hence rise in temperature will favour forward direction.

Example: In reaction $2H_2O(g) \rightleftharpoons 2H_2 + O_2(g)$; $\Delta H = +24.7$, which changes in condition will lead to maximum decomposition of $H_2O(g)$

Solution: Reaction is endothermic and $\Delta n > 0$, as hence the formation of the product will be favoured by increasing the temperature pressure.

Example: In reaction $Cl_2(g) + 3F_2(g) \rightleftharpoons 2ClF_3(g)$; $\Delta H = -329$ kJ

Which of following will increase the quantity of ClF_3

(A) Increasing the temperature

(B) Removing Cl_2

(C) Increasing the volume of container

(D) Adding F_2

Solution: (D) Equilibrium shift in forward direction when we add F_2

ELECTROLYTES AND NON-ELECTROLYTES

An electrolyte is defined as a compound whose aqueous solution or melt conducts electricity. On the other hand, a compound whose aqueous solution or melt does not conduct electricity is called a non-electrolyte.

e.g. Aqueous solution of sugar, urea etc. do not conduct electricity. Hence they are non- electrolytes whereas solutions of inorganic acids, bases and salts conduct electricity. Hence they are electrolytes.

(a) Types of electrolyte:

Electrolytes are of two types

(i) Weak electrolyte: A weak electrolyte is defined as a substance which dissociates to a small extent in aqueous solution and hence, conducts electricity to a small extent.

e.g. NH_4OH, CH_3COOH etc.

(ii) Strong electrolyte: A strong electrolyte is defined as a substance which is completely ionized in its aqueous solution and hence is a very good conductor of electricity.

e.g. NaOH, KOH, HCl, H_2SO_4, NaCl, KNO_3 etc.

IONIC EQUILIBRIUM

(a) Ionic Equilibrium

In case of weak electrolyte, as they are partly ionised in the aqueous solution, an equilibrium is setup between the ions and the unionised electrolyte.

Hence, their equilibrium is represented by putting double arrow ($\rightleftharpoons$) in between ionised and unionised form.

e.g. $CH_3COOH + H_2O \rightleftharpoons CH_3COO^-(aq) + H_3O^+$

In general, the ionisation of a weak electrolyte, AB, is represented as follows:

$AB(s) + H_2O \rightleftharpoons A^+(aq) + B^-(aq)$

Such an equilibrium between the ions and the undissociated electrolyte is called ionic equilibrium.

(b) Ionisation of Water

Pure water is a very weak electrolyte and ionised according to the following equation.

$H_2O(\ell) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$

Applying the law of mass action

$$K = \frac{[H_3O^+][OH^-]}{[H_2O]^2}$$

(H_2O) may be supposed to remain constant [$\because$ Water is ionised to a small extent.]

$\therefore K[H_2O]^2 = [H_3O^+][OH^-]$

$K_w = [H_3O^+][OH^-]$

$\therefore K_w$ = Constant (It is known as Ionic product of water).

Therefore the product of concentration of H^+ ions and OH^- ions in water at a particular temperature is known as ionic product of water.

The value of K_w increases with the increase of temperature, i.e. the concentration of H^+ and OH^- ions increase with increase in temperature.

Temperature	Value of K_w
at 0ºC	0.11×10^{-14}
at 10ºC	0.31×10^{-14}
at 25ºC	1.00×10^{-14}
at 100ºC	7.50×10^{-14}

(a) For neutral solution $[H_3O^+] = 10^{-7}$ M

(b) For acidic solution $[H_3O^+] > 10^{-7}$ M

(c) For basic solution $[H_3O^+] < 10^{-7}$ M

(c) Ostwald Dilution Law

(for weak electrolyte's)

- For a weak electrolyte A^+B^- dissolved is water, if α is the degree of dissociation then

$$AB \rightleftharpoons A^+ + B^-$$

initial conc	C	0	0
conc-at eq.	$C(1-\alpha)$	$C\alpha$	$C\alpha$

Then according to law of mass action,

$\frac{[A^+][B^-]}{[AB]} = \frac{C\alpha.C\alpha}{C(1-\alpha)} = \frac{C\alpha^2}{(1-\alpha)}$ = dissociation constant of the weak electrolyte

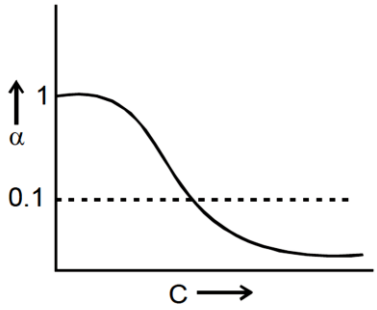
[$C = \frac{1}{V}$, then V = 1/C(volume of Solution in which 1 mole is present) is called **dilution**,

so $k_{eq} = \frac{\alpha^2}{(1-\alpha)V}$]

If α is negligible in comparison to unity then, $1 - \alpha \simeq 1$. So $K_{eq} = \alpha^2 C \Rightarrow \sqrt{\frac{k_{eq}}{c}} = \sqrt{k_{eq}.V}$.

$\alpha \propto \frac{1}{\text{concentration}}$ **{Thumb rule}**

- as **concentration increases $\Rightarrow$ α decreases**
- **at infinite dilution α reaches its maximum value, unity**.



(f) pH of a weak acid (monoprotic) Solution**:**

- Weak acid does not dissociate 100% therefore we have to calculate the percentage dissociation using K_a, dissociation constant of the acid.
- We have to use Ostwald's Dilution law (as have been derived earlier)

$$HA \rightleftharpoons H^+ + OH^-$$

t = 0	C	0	0
$t = t_{eq}$	$C(1-\alpha)$	$C\alpha$	$C\alpha$

$K_a = \frac{[H^+][OH^-]}{[HA]} = \frac{C\alpha^2}{1-\alpha}$

If $\alpha << 1 \Rightarrow (1-\alpha) \approx 1$ $\Rightarrow K_a \approx C\alpha^2$ $\Rightarrow \alpha = \sqrt{\frac{K_a}{C}}$ (is valid if $\alpha < 0.1$ or 10%)

$[H^+] = C\alpha = C\sqrt{\frac{K_a}{C}} = \sqrt{K_a \times C}$

So **pH** = $\frac{1}{2}(pK_a - \log C)$

on increasing the dilution

$\Rightarrow C \downarrow$ $\Rightarrow \alpha \uparrow$ and $[H^+] \downarrow \Rightarrow$ pH $\uparrow$

Example: Calculate pH of

(a) 10^{-1} M CH_3COOH

(b) 10^{-3} M CH_3COOH

(c) 10^{-6} M CH_3COOH

Take $K_a = 2 \times 10^{-5}$.

Solution: **(a)** $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

C	0	0
C(1–α)	Cα	Cα

$K_a = \frac{C\alpha^2}{1-\alpha} \Rightarrow \alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{2\times10^{-5}}{10^{-1}}} = \sqrt{2\times10^{-4}}$ $(\alpha << 0.1)$

So, $[H^+] = 10^{-1} \times \sqrt{2} \times 10^{-2} \Rightarrow pH = 3 - \frac{1}{2}\log 2 = 2.85$ **Ans.**

(b) $\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{2\times10^{-5}}{10^{-3}}} = \sqrt{2\times10^{-2}}$ $(\alpha > 0.1)$

So we have to do the exact calculations

$K_a = \frac{C\alpha^2}{1-\alpha} \Rightarrow 2\times10^{-5} = \frac{10^{-3}\times\alpha^2}{1-\alpha} \Rightarrow \alpha = 13.14\%$

$[H^+] = 10^{-3} \times 0.1314 = 1.314 \times 10^{-4} \Rightarrow pH = 4 - \log(1.314) \approx 3.8$ **Ans.**

(c) If approximation is used then, $\alpha = \sqrt{\frac{2\times10^{-5}}{10^{-6}}} = \sqrt{20} > 1$,

so we have to do the exact calculation, $2\times10^{-5} = 10^{-6}\frac{\alpha^2}{1-\alpha} \Rightarrow \alpha \approx 0.95$ or 95%

$[H^+] = 0.95 \times 10^{-6} = 9.5 \times 10^{-7} \Rightarrow pH = 7 - \log(9.5) = 6.022$ **Ans.**

◈ **At very low concentration (at infinite dilution) weak electrolyte will be almost 100% dissociated, so behave as strong electrolyte**

(pH) of 10^{-6} M HCl $\simeq$ pH of 10^{-6} M $CH_3COOH \simeq 6$)

Example: A 0.01 M solution of acetic acid is 5% ionised at 25°C. Calculate its dissociation constant.

Solution: According to Ostwald's dilution law,

$K_a = \frac{\alpha^2}{(1-\alpha)V}$

$\alpha = 0.05$, $V = \frac{1}{0.01} = 100$ litres

Hence, $K_a = \frac{0.05\times0.05}{(1-0.05)100} = 2.63 \times 10^{-5}$

Example: Calculate the H^+ ion concentration of a 0.01 N weak monobasic acid. The value of dissociation contant is 4.0×10^{-10}.

Solution: $HA \rightleftharpoons H^+ + A^-$

Applying Ostwald's dilution law for a weak acid,

$\alpha = \sqrt{K_a V}$

$K_a = 4.0 \times 10^{-10}$, $V = \frac{1}{0.01} = 100$ litres

$a = \sqrt{4.0\times10^{-10}\times10^{2}} = 2 \times 10^{-4}$

concentration of hydrogen ions = $\frac{2\times10^{-4}}{100} = 2 \times 10^{-6}$ mol L^{-1}

or concentration of hydrogen ions = $\sqrt{CK} = \sqrt{0.01\times4\times10^{-10}} = 2 \times 10^{-6}$ mol L^{-1}

Example: What will be the dissociation constant of 0.1 N aqueous ammonia solution in terms of degree of dissociation 'α'? What will be the value if the concentration is 0.01 N?

Solution: $NH_4OH \rightleftharpoons NH^+_4 + OH^-$

At equilibrium $(1-\alpha)$ α α

$V = \frac{1}{0.1} = 10$ litre

$[NH_4OH] = \frac{(1-\alpha)}{10}$, $[NH_4^+] = \frac{\alpha}{10}$ and $[OH^-] = \frac{\alpha}{10}$

Applying law of mass action,

$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]} = \frac{\frac{\alpha}{10}\times\frac{\alpha}{10}}{\frac{1-\alpha}{10}} = \frac{\alpha^2}{10(1-\alpha)}$

For 0.01 N solution, K_b remains the same at the same temperature but degree of dissociation value becomes different.

SALT HYDROLYSIS

(Reverse of neutralisation)

Salt + Water $\rightleftharpoons$ acid +base

When acids and bases are mixed in equal amounts (equivalents), then we will be having salt Solutions only in the water and we have to calculate pH of salt Solutions.

When a salt is added to water, the solid salt first dissolves and breaks into ions completely (unless otherwise specified). The ions of the salt may or may not react with water the cations when react with water will always produce H_3O^+ ions and the anions on reaction with water will produce OH^- ions. Depending on the extent of hydrolysis and on the amounts of H_3O^+ and OH^- ions the Solution can be acidic, basic or neutral. If salt is BA, then

$BA(s) \longrightarrow BA(aq) \longrightarrow B^+(aq) + A^-(aq)$

$A^-(aq) + H_2O(l) \rightleftharpoons HA(aq) + OH^-(aq)$ (anionic hydrolysis)

$B^+(aq) + 2H_2O(l) \rightleftharpoons BOH(aq) + H_3O^+(aq)$ (cationic hydrolysis)

ANIONIC HYDROLYSIS

Anions can function as a base on reaction with water and hydrolyse as follows:

$A^-(aq) + H_2O(l) \rightleftharpoons HA\ (aq) + OH\ (aq)$

The extent of hydrolysis of a given anion depends on its basic strength

CATIONIC HYDROLYSIS

Cations can function as acid on reaction with water and hydrolyse as follows.

$B^+(aq) + 2H_2O(l) \rightleftharpoons BOH(aq)\ H_3O^+(aq)$

The extent of hydrolysis of a given cation depends on its acidic strength.

There are four types of salt.

(i) Salt of strong acid and strong base
(ii) Salt of strong acid and weak base
(iii) Salt of weak acid and strong base
(iv) Salt of weak acid and weak base

Salts of first type does not undergo hydrolysis and rest three types undergo hydrolysis.

(i) Salt of strong acid and strong base:

Neither of the ions will undergo hydrolysis so the Solution contain only the equilibrium of ionization of water.

$2H_2O(l) \rightleftharpoons H_3O^+ + OH^-$

Thus, **the pH of** Solution **will be 7(neutral** Solution**).**

(ii) Salt of strong acid and weak base:

The ***Example***s can be NH_4Cl, $(NH_4)_2\ SO_4$, $AlCl_3$

Only the cation will undergo hydrolysis and the Solution will be acidic in nature. for ***Example*** in the solution of NH_4Cl of concentration c, we will have

$$NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$$

t = 0	c	0	0	0
at eq.	c(1– h)		ch	ch

(h - degree of hydrolysis)

$K_h = \dfrac{[NH_4OH][H^+]}{[NH_4^+]}$ = called hydrolysis constant of the salt

$NH_4OH \rightleftharpoons NH_4^+ + OH^-$, $K_b = \dfrac{[NH_4^+][OH^-]}{[NH_4OH]}$

$H_2O \rightleftharpoons H^+ + OH^-$, $K_w = [H^+]\ [OH^-]$

From above equations we can get

$K_h \times K_b = K_w$

$K_h = \dfrac{ch.ch}{c(1-h)} = \dfrac{ch^2}{(1-h)}$ (generally 1-h $\simeq$ 1) so we get, $\Rightarrow h = \sqrt{\dfrac{K_h}{c}}$

$\Rightarrow [H^+] = ch = \sqrt{K_h \times c} = \sqrt{\dfrac{K_w}{K_b} \times c}$

$\Rightarrow pH = -\log [H^+] = -\dfrac{1}{2}[\log K_w - \log K_b + \log c] = \dfrac{1}{2}$ **$[pK_w - pK_b - \log c]$**

(iii) Salt of weak acid and strong base:

The ***Example***s can be CH_3COONa, KCN, $Na_2C_2O_4$, K_3PO_4

Similar to above analysis we will get

$k_h = \frac{ch.ch}{c(1-h)} = \frac{ch^2}{(1-h)}$ (generally 1-h $\simeq$ 1) so we get, $\Rightarrow h = \sqrt{\frac{k_h}{c}}$

$k_h \times k_a = k_w$

$[OH^-] = ch = \sqrt{k_h \times c} = \sqrt{\frac{k_w}{k_a} \times c}$

$pH = -\log [H^+] = -\frac{1}{2} [\log k_w + \log k_a + \log c]$

$= \frac{1}{2}$ **$[pK_w + pK_a + \log c]$**

(iv) Salt of weak acid and weak base:

***Example*s** can include CH_3COONH_4, NH_4CN, $Ca(CH_3COO)_2$, MgC_2O_4

$$CH_3COO^- + NH_4^+ + H_2O \rightleftharpoons CH_3COOH + NH_4OH$$

t = 0	c	c	0	0
at eq.	c – ch	c – ch	ch	ch

$K_h = \frac{[CH_3COOH][NH_4OH]}{[CH_3COO^-][NH_4^+]}$ .....(i)

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$,

$Ka = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$ .....(ii)

$NH_4OH \rightleftharpoons NH_4^+ + OH^-$,

$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]}$ .....(iii)

$H_2O \rightleftharpoons H^+ + OH^-$,

$K_w = [H^+]\,[OH^-]$ .....(iv)

So, **$K_h \times K_a \times K_b = K_w$,**

$\Rightarrow k_h = \frac{ch.ch}{c(1-h).c(1-h)} = \left(\frac{h}{1-h}\right)^2$

$\Rightarrow \left(\frac{h}{1-h}\right) = \sqrt{K_h}$

from (ii) equation

$[H^+] = k_a \frac{[CH_3COOH]}{[CH_3COO^-]} = K_a \frac{ch}{c(1-h)} = K_a \times \frac{h}{1-h} = K_a \times \sqrt{K_h} = K_a \times \sqrt{\frac{K_w}{K_a \times K_b}} = \sqrt{\frac{K_w \times K_a}{K_b}}$

pH $= -\log [H^+] = \frac{1}{2}$ **$[pK_w + pK_a - pK_b]$**

Example: The degree of hydrolysis of which of the following salts is independent of the concentration of salt solution?

(A) CH_3COONa (B) NH_4Cl (C) CH_3COONH_4 (D) NaCl

Solution: **(C)**

It is the salt of weak acid and weak base, hence its degree of hydrolysis will be independent of concentration.

$$h = \sqrt{\frac{K_w}{K_a \times K_b}}$$

Example: The pH of 1 M PO_4^{3-} (aq) solution will be (given pK_b of PO_4^{3-} = 1.62):

(A) 1.62 (B) 12.38 (C) 13.19 (D) 7

Solution: **(C)**

$pK_a = 14 - pK_b = 14 - 1.62 = 12.38$

$$pH = \frac{1}{2}[pK_w + pK_b + \log C]$$

$$\Rightarrow \frac{1}{2}[14 + 12.38 + 0] = 13.19$$

Example: For the following equilibrium

$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

Calculate the equilibrium constant, if for the equilibrium,

$NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$

the equilibrium constant is 5.5×10^{-10}

(A) 1.8×10^{-4} (B) 1.8×10^{-5} (C) 1.8×10^{-6} (D) 1.8×10^{-7}

Solution: **(B)**

$K_w = K_a \times K_b$

$10^{-14} = 5.5 \times 10^{-10} \times K_b$

$K_b = 1.8 \times 10^{-5}$

COMMON ION EFFECT & SOLUBILITY PRODUCT

COMMON ION EFFECT

Let AB be a weak electrolyte. Considering its dissociation,

$AB \rightleftharpoons A^+ + B^-$.

and applying law of mass action we get

$$K = \frac{[A^+][B^-]}{[AB]}$$

The equilibrium constant 'K' has a definite value at any given temperature. If now another electrolyte furnishing the A^+ or B^- ions is added to the above solution, it will increase the concentration of either A^+ ions or B^- ion (whichever has been added) and in order to keep K constant, the concentration of AB must increase, i.e, the equilibrium will shift to the left hand side.

In other words, the degree of dissociation of an electrolyte (weak) is suppressed by the addition of another electrolyte (strong) containing a common ion. This is termed as common ion effect.

e.g. When sodium acetate (CH_3COONa) is added to the solution of acetic acid (CH_3COOH), then the degree of dissociation of acetic acid (weak acid) is suppressed due to common ion effect.

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ ... (i)

Weak electrolyte

$CH_3COONa \longrightarrow CH_3COO^- + Na^+$ ... (ii)

Strong electrolyte Common acetate ion

Applying law of mass action on equilibrium.

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} \quad \text{... (iii)}$$

$$[H^+] = \frac{K_a[CH_3COOH]}{[CH_3COO^-]} \quad \text{... (iv)}$$

Above relation shows, that on increasing the concentration of acetate ion (CH_3COO^-), hydrogen ion concentration decreases which shows that, in presence of sodium acetate degree of ionization of acetic acid is decreased.

SOLUBILITY PRODUCT

In a saturated solution of an electrolyte two equilibrium exist and can be represented as:

$$AB \rightleftharpoons AB \rightleftharpoons \underbrace{A^+ + B^-}_{\text{Ions}}$$

Solid Unionised (dissolved)

Applying the law of mass action to the ionic equilibrium,

$$\frac{[A^+][B^-]}{[AB]} = K$$

Since the solution is saturated, the concentration of unionised molecules of the electrolyte is constant at a particular temperature.

i.e. $[AB] = K' = \text{constant}$.

$[A^+][B^-] = K \times K' = K_{SP}$

Where K_{SP} is termed as the solubility product. It is defined as the product of the concentration of ions in a saturated solution of an electrolyte at a given temperature.

Example: If solubility of $CaCO_3$ in water is 0.0305 g/L. What will be the solubility product of $CaCO_3$.

Solution: Solubility in mole/litre (S) $= \frac{\text{Solubility in g / L}}{\text{Molecular weight}} = \frac{0.0305}{100} = 3.05 \times 10^{-4}$

Since dissolved $CaCO_3$ is completely ionized -

$CaCO_3 \rightleftharpoons Ca^{2+} + CO_3^{2-}$

$K_{SP} = [Ca^{2+}][CO_3^{2-}]$

$K_{SP} = (3.05 \times 10^{-4}) \times (3.05 \times 10^{-4}) = 9.3 \times 10^{-8}$

Example: Solubility of CaF_2 in water is 0.0002 mol/litre. Calculate its solubility product.

Solution: Solubility of CaF_2 = 0.0002 mol/litre = 2×10^{-4} mol/litre.

$$CaF_2 \rightleftharpoons Ca^{2+} + 2F^-$$

(S) (2S)

molar concentration of F^- will be twice of Ca^{2+}

i.e. $[Ca^{2+}] = 2 \times 10^{-4}$

$[F^-] = 2 \times 2 \times 10^{-4}$

$K_{SP} = [Ca^{2+}] [F^-]^2$

$= (2 \times 10^{-4}) \times (2 \times 2 \times 10^{-4})^2$

$= 32 \times 10^{-12}$

$= 3.2 \times 10^{-11}$

INDICATORS & ACID BASE TITRATION CURVES:

THEORY OF INDICATORS

An indicator is a substance which is used to determine the end point in a titration. In acid-base titrations organic substances (weak acids or weak bases) are generally used as indicators. They change their colour within a certain pH range. The colour change and the pH range of some common indicators are tabulated below:

Indicator	pH range	Colour change
Methyl orange	3.2 – 4.5	Red to orange
Methyl red	4.4 – 6.5	Red to yellow
Litmus	5.5 – 7.5	Red to blue
Phenol red	6.8 – 8.4	Yellow to red
Phenolphthalein	8.3 – 10.5	Colourless to pink

Theory of acid-base indicators: Two theories have been proposed to explain the change of colour of acid-base indicators with change in pH.

1. Ostwald's theory:

According to this theory

(a) The colour change is due to ionisation of the acid-base indicator. The unionised form has different colour than the ionised form.

(b) The ionisation of the indicator is largely affected in acids and bases as it is either a weak acid or a weak base. In case, the indicator is a weak acid, its ionisation is very much low in acids due to common ions while it is fairly ionised in alkalies. Similarly if the indicator is a weak base, its ionisation is large in acids and low in alkalies due to common ions.

Considering two important indicators phenolphthalein (a weak acid) and methyl orange (a weak base), Ostwald's theory can be illustrated as follows:

Phenolphthalein: It can be represented as HPh. It ionises in solution to a small extent as:

$$HPh \rightleftharpoons H^+ + Ph^-$$

Colourless Pink

Applying law of mass action,

$$K = \frac{[H^+][Ph^-]}{[HPh]}$$

The undissociated molecules of phenolphthalein are colourless while ph^- ions are pink in colour. In presence of an acid, the ionisation of HPh is practically negligible as the equilibrium shifts to left hand side due to high concentration of H^+ ions. Thus, the solution would remain colourless. On addition of alkali, hydrogen ions are removed by OH^- ions in the form of water molecules and the equilibrium shifts to right hand side. Thus, the concentration of ph^- ions increases in solution and they impart pink colour to the solution.

Let us derive Henderson's equation for an indicator

$$Hln + H_2O \rightleftharpoons H_3^+O + ln^-$$

'Acid form' 'Base form'

Conjugate acid-base pair

$$K_{In} = \frac{[In^-][H_3^+O]}{[HIn]}$$

K_{In} = Ionization constant of indicator

$$[H_3^+O] = K_{In} \times \frac{[HIn]}{[In^-]}$$

$$pH = -\log_{10}[H_3^+O] = -\log_{10}[K_{In}] - \log_{10}\frac{[HIn]}{[In^-]}$$

$$pH = pK_{In} + \log_{10}\frac{[In^-]}{[HIn]}$$ (Henderson's equation for indicator)

At equivalence point ;

$[In^-] = [HIn]$ and $pH = pK_{In}$

Methyl orange: It is a very weak base and can be represented as MeOH. It is ionised in solution to give Me^+ and OH^- ions.

$$MeOH \rightleftharpoons Me^+ + OH^-$$

Orange Red

Applying law of mass action,

$$K = \frac{[Me^+][OH^-]}{[MeOH]}$$

In presence of an acid, OH^- ions are removed in the form of water molecules and the above equilibrium shifts to right hand side. Thus, sufficient Me^+ ions are produced which impart red colour to the solution. On addition of alkali, the concentration of OH^- ions increases in the solution and the equilibrium shifts to left hand side, i.e., the ionisation of MeOH is practically negligible. Thus, the solution acquires the colour of unionised methyl orange molecules, i.e. orange.

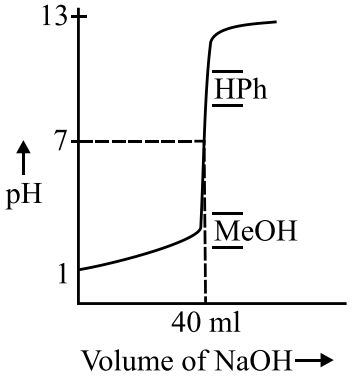
This theory also explains the reason why phenolphthalein is not a suitable indicator for titrating a weak base against strong acid. The OH^- ions furnished by a weak base are not sufficient to shift the equilibrium towards right hand side considerably, i.e., pH is not reached to 8.3. Thus, the solution does not attain pink colour. Similarly, it can be explained why methyl orange is not a suitable indicator for the titration of weak acid with strong base.

ACID-BASE TITRATION:

- Titration of SA vs SB:

40 ml 0.1 M HCl, NaOH 0.1 M

Volume of NaOH added	**pH of Solution**
0 ml	1
10 ml	1.22
20 ml	1.48
30 ml	1.84
39 ml	2.90
39.9 ml	3.90
40.0 ml	7
40.1 ml	10.1
41 ml	11.1
50 ml	12.05
60 ml	12.30



⇒ Indicator should change it colour sharply for indication of reaction to be complete so its pH- range should lie perfectly with in sharp change to avoid experimental error.

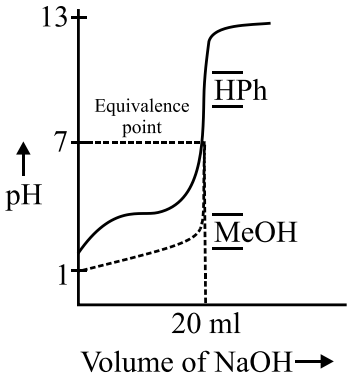
MeOH and HpH both can be used for titration of SA and SB.

- Titration of WA vs SB:

WA = CH_3COOH, Ka = 2×10^{-5},

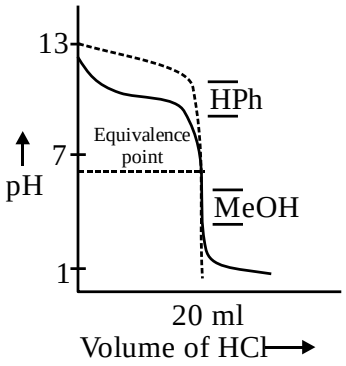
0.5, 20 ml, NaOH = 0.5 M

Volume of NaOH	pH value
0	2.5
5 ml	4.23
10 ml	4.7
15 ml	4.7 + log 3 = 5.17
19 ml	4.7 + log 19/1 = 6
20 ml	7 + 1/2 (4.7) + 1/2 log (1/4) = 9.05
21 ml	12.4
30 ml	13.23



* In sharp change of pH HPh is suitable indicator.
* Initially fast change in pH is due to free ions available from weak acid.
* Slow down of pH change is due to Buffer formation, which is maximum at pH = pKa.
* Near equivalence point pH again increases fastly due to salt formation.
* After equiv. point pH changes very sharply due to presence of strong base.

- Titration of WB vs SA



Sharp change between 4 to 7

EXERCISE-1

1. Which of the following reactions is an endothermic reaction?

(A) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g) - Heat$

(B) $2H_2O \xrightarrow{\text{Electric current}} 2H_2(g) + O_2(g)$

(C) $N_2(g) + O_2(g) \longrightarrow 2NO(g) - Heat$

(D) All are endothermic reactions

2. Which of the following does not disturb the equilibrium point?

(A) Concentration (B) Temperature

(C) Catalyst (D) Pressure

3. Which of the following statements is not correct?

(A) Combustion reactions are exothermic.

(B) Most of the reactions are endothermic.

(C) Decomposition reactions are generally endothermic.

(D) In exothermic reactions potential energy of product is less than that of reactants.

4. A reversible reaction is the one in which -

(A) the reactants are completely converted into products.

(B) half of the reactants are used up.

(C) the rates of both the forward & the reverse reactions are unequal.

(D) the concentrations of reactants & products become fixed at equilibrium.

5. The reaction $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$ is irreversible only when -

(A) the reaction is carried out in a closed vessel

(B) the reaction is carried out in an open vessel

(C) the reaction takes place at a definite temperature & pressure

(D) none of these

6. For an equilibrium reaction $H_2 + I_2 \rightleftharpoons 2HI$ Value of Δn will be -

(A) 2 (B) 1

(C) zero (D) 3

7. A chemical reaction $A \rightleftharpoons B$ is said to be at equilibrium when:

(A) Complete conversion of A to B has taken place

(B) Conversion of A to B is only 50% complete

(C) Only 10% conversion of A to B has taken place

(D) The rate of transformation of A to B is just equal to the rate of transformation of B to A in the system.

8. For $N_2 + 3H_2 \rightleftharpoons 2\,NH_3$; ΔH = – ve, then:

(A) $K_p = K_c$ (B) $K_p = K_cRT$

(C) $K_p = K_c(RT)^{-2}$ (D) $K_p = K_c(RT)^{-1}$

9. For the reaction, $H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$ equilibrium constant, K_p changes with:

(A) Total pressure

(B) Catalyst

(C) Amount of H_2 and I_2 present

(D) Temperature

10. For the reaction $H_2 + I_2 \rightleftharpoons 2HI$:

(A) $K_c = 2K_p$ (B) $K_c > K_p$

(C) $K_c = K_p$ (D) $K_c < K_p$

11. The equilibrium constant for the reaction $mA + nB \rightleftharpoons pC + qD$ may be written as -

(A) $K_C = \dfrac{[A]^m[B]^n}{[C]^p[D]^q}$

(B) $K_C = \dfrac{[AB]^{m+n}}{[CD]^{p+q}}$

(C) $K_C = \dfrac{[C]^p[D]^q}{[A]^m[B]^n}$

(D) $K_C = \dfrac{[C]^{1/m}[D]^{1/n}}{[A]^{1/p}[B]^{1/q}}$

12. The equilibrium constant for the following reaction will be

$$3\,A + 2\,B \rightleftharpoons C$$

(A) $\frac{[3A][2B]}{[C]}$ (B) $\frac{[C]}{[3A][2B]}$

(C) $\frac{[C]}{[A]^2[B]^2}$ (D) $\frac{[C]}{[A]^3[B]^2}$

13. The rate constant of forward reaction is two times that of the reverse reaction at a given temperature, the value of equilibrium constant is -

(A) 0·5 (B) 1·5

(C) 2·5 (D) 2·0

14. If K_1 and K_2 are the equilibrium constants of the equilibria (a) and (b) respectively, what is the relationship between the two constants?

(a) $SO_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons SO_3(g)$; K_1

(b) $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$; K_2

(A) $(K_1)^2 = \frac{1}{K_2}$ (B) $K_2 = (K_1)^2$

(C) $K_1 = \frac{1}{K_2}$ (D) $K_1 = K_2$

15. At a given temperature, the equilibrium constant for the reactions

$NO(g) + 1/2O_2(g) \rightleftharpoons NO_2(g)$ and $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$ are K_1 and K_2 respectively. If K_1 is 4×10^{-3} then K_2 will be

(A) 8×10^{-3} (B) 16×10^{-3}

(C) $6{\cdot}25 \times 10^{4}$ (D) $6{\cdot}25 \times 10^{6}$

16. The equilibrium concentration of x, y and yx_2 are 4, 2 and 2 respectively for the equilibrium $2x + y \rightleftharpoons yx_2$. The value of equilibrium constant, K_c is

(A) 0.625 (B) 6.25

(C) 0.0625 (D) 62.5

17. The extent of dissociation of PCl_5 at a certain temperature is 20 % at one atm pressure. Calculate the approx pressure at which this substance is half dissociated at the same temperature.

(A) 0.123 (B) 0.246

(C) 0.826 (D) 0.111

18. HI was heated in a sealed tube at 400°C till the equilibrium was reached. HI was found to be 22% decomposed. The equilibrium constant for dissociation is:

(A) 1.99 (B) 0.0199

(C) 0.0796 (D) 0.282

19. A reaction in equilibrium is represented by the following equation –

$2A(s) + 3B(g) \rightleftharpoons 3C(g) + D(g) + O_2$ if the pressure on the system is reduced to half of its original value

(A) The amounts of C and D decreases

(B) The amounts of C and D increases

(C) The amount of B and D decreases

(D) All the amounts remain constant

20. In which of the following equilibrium reactions, the equilibrium would shift to right side, if total pressure is decreased:

(A) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

(B) $H_2 + I_2 \rightleftharpoons 2HI$

(C) $N_2O_4 \rightleftharpoons 2NO_2$

(D) $H_2 + Cl_2 \rightleftharpoons 2HCl$

21. The effect of increasing the pressure on the following equilibrium $2A + 3B \rightleftharpoons 3A + 2B$ is -

(A) Forward reaction is favoured

(B) Backward reaction is favoured

(C) No effect

(D) None of these

22. For the system $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$ if the concentration of NH_3 is doubled, the equilibrium constant will -
(A) be doubled
(B) increase, but by less than a factor of 2
(C) be halved
(D) remains the same

23. The value of equilibrium constant for an unfavourable forward reaction is -
(A) less than one (B) equal to one
(C) more than one (D) None of these

24. Consider the following reaction in a closed container
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$.
At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant (K_p) and degree of dissociation (α)
(A) neither K_p nor α changes
(B) both K_p and α change
(C) K_p changes, but α does not change
(D) K_p does not change, but α changes

25. The value of ionic product of water at 25ºC is -
(A) 1×10^{-1} mol^{2}L^{-2}
(B) 1×10^{-14} mol^{2}L^{-2}
(C) 1×10^{14} mol^{2}L^{-2}
(D) 1×10^{7} mol^{2}L^{-2}

26. In a chemical system at equilibrium, the addition of catalyst would result in -
(A) Increase in rate of forward reaction only
(B) Increase in rate of backward reaction only
(C) No effect on rate of reaction
(D) Provide a new path having low activation energy of the reaction

27. Aniline is a very weak base. Which of the given aniline solution will have highest degree of dissociation?
(A) 1M aniline (B) 0.1 M aniline
(C) 0.01 M aniline (D) 0.02 M aniline

28. The pH of a 0.1 M solution of weak base is 11. On diluting the solution:
(A) pH increases (B) $[OH^-]$ increases
(C) α decreases (D) pH decreases

29. A monoprotic acid in a 0.1 M solution ionize to 0.001%. Its ionization constant is –
(A) 1.0×10^{-3} (B) 1.0×10^{-6}
(C) 1.0×10^{-8} (D) 1.0×10^{-11}

30. Ka for a weak acid HA is 1.44×10^{-5}. What is the concentration of A^- when 0.01 mole of HA is dissolved in 1L of 0.01M HCl solution?
(A) 0.01 M (B) 1.2×10^{-3} M
(C) 1.44×10^{-5} M (D) 0.012 M

31. Which of the following reactions is an endothermic one?
(A) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(B) $C(s) + O_2(g) \rightleftharpoons CO_2(g)$
(C) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$(D) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

32. In which of the following reaction rate increases with the increase of temperature?
(A) Endothermic reactions
(B) Exothermic reactions
(C) Any reaction
(D) None

33. When H_2 and I_2 are mixed and equilibrium is attained, then -
(A) amount of HI formed is equal to the amount of H_2 dissociated
(B) HI dissociation stops
(C) the reaction stops completely
(D) None of these

34. In the reaction, $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$:
(A) $K_P = K_C$ (B) $K_P \neq K_C$
(C) $K_P > K_C$ (D) $K_P < K_C$

35. If 0.5 mole H_2 is reacted with 0.5 mole I_2 in a ten-litre container at 444°C and at same temperature value of equilibrium constant K_C is 49, the ratio of [H] and [I_2] will be:

(A) 7 (B) $\frac{1}{7}$

(C) $\sqrt{\frac{1}{7}}$ (D) 49

36. What should be the relationship between K_2 and K_1, if the equilibrium constant of the reaction given below are K_1 and K_2 respectively?

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $SO_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons SO_3(g)$

(A) $K_2 = K_1$ (B) $K_2 = \sqrt{K_1}$

(C) $K_2 = K_1$ (D) $2K_2 = K_1$

37. The correct expression for the equilibrium constant for the reaction - $NH_3(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g)$ is

(A) $\frac{[N_2]^{1/2}[H_2]^{3/2}}{[NH_3]}$ (B) $\frac{[N_2][H_2]}{[NH_3]}$

(C) $[N_2]\,[H_2] - [NH_3]$ (D) $\frac{[N_2]^{1/2}[H_2]^{3/2}}{[NH_3]^2}$

38. If K_p for a reaction

$A(g) + 2B(g) \rightleftharpoons 3C(g) + D(g)$

is 0·05 atm at 1000 K. Its K_c in terms of R will be

(A) 20000 R (B) 0·02 R

(C) 5×10^{-5} R (D) $\frac{5\times10^{-5}}{R}$

39. 4 moles of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 calculate total number of moles at equilibrium:

(A) 4.5 (B) 6

(C) 3 (D) 4

40. In a 0.25 litre tube dissociation of 4 moles of NO is take place. If its degree of dissociation is 10%.

The value of K_p for reaction $2NO \rightleftharpoons N_2 + O_2$ is:

(A) $\frac{1}{(18)^2}$ (B) $\frac{1}{(8)^2}$

(C) $\frac{1}{16}$ (D) $\frac{1}{32}$

41. At constant temperature, the equilibrium constant (K_p) for the decomposition reaction $N_2O_4 \rightleftharpoons 2NO_2$ is expressed by $K_p = 4x^2 P/(1–x^2)$, where P is pressure, x is extent of decomposition. Which of the following statement is true?

(A) K_p increases with increase of P
(B) K_p increases with increase of x
(C) K_p increases with decrease of x
(D) K_p remains constant with change in P or x

42. For the equilibrium $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + 92$ kJ. The ammonia formation is favoured by

(A) increasing the pressure
(B) Adding more nitrogen
(C) Decreasing the temperature
(D) All

43. In the given reaction:

$2SO_2(g) + O_2 \rightleftharpoons 2SO_3(g) + \text{heat}$

The formation of sulphur trioxide is favoured by -

(A) increasing the pressure
(B) decreasing the temperature
(C) adding more amount of oxygen
(D) all of the above

44. For the chemical reaction

$3X(g) + Y(g) \rightleftharpoons X_3Y(g)$

the amount of X_3Y at equilibrium is affected by

(A) temperature and pressure
(B) temperature only
(C) pressure only
(D) temperature, pressure and catalyst.

45. The oxidation of SO_2 by O_2 to SO_3 is an exothermic reaction. The yield of SO_3 will be maximum if

(A) temperature is increased and pressure is kept constant

(B) temperature is reduced and pressure is increased

(C) both temperature and pressure are increased

(D) both temperature and pressure are decreased

46. In which of the following equilibrium, change in the volume of the system does not alter the number of moles.

(A) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

(B) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(C) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(D) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$.

47. Consider the reaction

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

in closed container at equilibrium. What would be the effect of addition of $CaCO_3$ on the equilibrium concentration of CO_2?

(A) Increases

(B) Decreases

(C) Data is not sufficient to predict it

(D) Remains unaffected

48. When pressure is applied to the equilibrium system, ice $\rightleftharpoons$ water,

which of the following phenomenon will occur?

(A) More ice will be formed

(B) Water will be evaporated

(C) More water will be formed

(D) Equilibrium will not be disturbed

49. The addition of HCl will not suppress the ionisation of -

(A) acetic acid

(B) sulphuric acid

(C) H_2S

(D) benzoic acid

50. On addition of ammonium chloride to a solution of NH_4OH -

(A) dissociation of NH_4OH increases

(B) concentration of OH^- decreases

(C) concentration of OH^- increases

(D) concentration of both NH_4^+ and OH^- increases

51. The following equilibrium exists in aqueous solution -

$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$

If dilute HCl is added –

(A) the equilibrium constant will increase

(B) the equilibrium constant will decrease

(C) unionised acetic acid concentration will increase

(D) acetate ion concentration will not change

52. The equilibrium constant, K_p for the reaction

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

is 4.0 atm^{-1} at 1000 K. What would be the partial pressure of O_2 if at equilibrium the amount of SO_2 and SO_3 is the same?

(A) 16.0 atm

(B) 0.25 atm

(C) 1 atm

(D) 0.75 atm

53. For the reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of A is a and that of B is 1.5 times that of A. Concentration of A and D are same at equilibrium. What should be the concentration of B at equilibrium?

(A) $\frac{a}{4}$

(B) $\frac{a}{2}$

(C) $\frac{3a}{4}$

(D) All of the above

54. One mole PCl_5 is heated in a closed container of one litre capacity. At equilibrium 20% PCl_5 is not dissociated. What should be the value of K_C?

(A) $(3 - 2)^{-1}$

(B) 3.2

(C) 2.4

(D) 42

55. How many moles per litre of PCl_5 has to be taken to obtain 0.1 mole Cl_2, if the value of equilibrium contant K_C is 0.04?

(A) 1.15 (B) 0.25

(C) 0.35 (D) 0.05

56. At 1000 K, the value of K_p for the reaction:

$A(g) + 2B(g) \rightleftharpoons 3C(g) + D(g)$ is 0.05 atm. The value of K_C in terms of R would be:

(A) 20000 R

(B) 0.02 R

(C) 5×10^{-5} R

(D) $5 \times 10^{-5} \times R^{-1}$

57. 4 moles of A are mixed with 4 moles of B, when 2 moles of C are formed at equilibrium, according to the reaction, $A + B \rightleftharpoons C + D$. The equilibrium constant is:

(A) 4 (B) 1

(C) 2 (D) 4

58. In the reaction $2P(g) + Q(g) \rightleftharpoons 3R(g) + S(g)$. If 2 moles each of P and Q taken initially in a 1 litre flask. At equilibrium which is true:

(A) [P] < [Q]

(B) [P] = [Q]

(C) [Q] = [R]

(D) None of these

59. Ammonia dissociates into N_2 and H_2 such that degree of dissociation α is very less than 1 and equilibrium pressure is P_0 then the value of α is [If K_p for $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ is $27 \times 10^{-8} P_0^2$

(A) 10^{-4}

(B) 4×10^{-4}

(C) 0.02

(D) can't be calculated

60. At a temp, T, a compound $AB_4(g)$ dissociates as $2AB_4(g) \rightleftharpoons A_2(g) + 4B_2(g)$ with a degree of dissociation x, which is small compared with unity. The expression of Kp in terms of x and total pressure P is:

(A) $8P^3x^5$

(B) $256P^3x^5$

(C) $4Px^2$

(D) None of these

61. The vapour density of N_2O_4 at a certain temperature is 30. What is the percentage dissociation of N_2O_4 at this temperature?

(A) 53.3 %

(B) 106.6%

(C) 26.7%

(D) None

62. At temperature T, a compound $AB_2(g)$ dissociates according to the reaction:

$2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$

with a degree of dissociation 'x' which is small compared to unity. The expression for 'x' in terms of the equilibrium constant K_P and the total pressure P is:

(A) $(K_P/P)^{1/3}$

(B) $(K_P/2P)^{1/3}$

(C) $(2K_P/P)^{1/3}$

(D) $(2K_P/3P)^{1/3}$

EXERCISE-2

1. Two solids dissociate as follows,

$A(s) \rightarrow B(g) + C(g)$; $K_{P_1} = x\ atm^2$

$D(s) \rightarrow C(g) + E(g)$; $K_{P_2} = y\ atm^2$

The total pressure when both the solids dissociate simultaneously is:

(A) $\sqrt{x+y}$ atm (B) $2(\sqrt{x+y})$atm

(C) $(x+y)$ atm (D) x^2+y^2 atm

2. For the following reactions, equilibrium constants are given:

$S(s) + O_2(g) \rightleftharpoons SO_2(g)$; $K_1 = 10^{52}$;

$2S(s) + 3O_2(g) \rightleftharpoons 2SO_3(g)$; $K_2 = 10^{129}$

The equilibrium constant for the reaction, $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is:

(A) 10^{154} (B) 10^{181}

(C) 10^{25} (D) 10^{77}

3. In which one of the following equilibria, $K_p \neq K_C$?

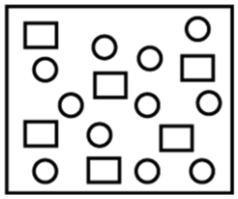
(A) $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$

(B) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

(C) $NO_2(g) + SO_2(g) \rightleftharpoons NO(g) + SO_3(g)$

(D) $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$

4. In the figure shown below, reactant A (represented by square) is in equilibrium with product B (represented by circle). The equilibrium constant is:



(A) 4 (B) 8

(C) 1 (D) 2

5. If the equilibrium constant for $A \rightleftharpoons B + C$ is $K_{eq}^{(1)}$ and that of $B + C \rightleftharpoons P$ is $K_{eq}^{(2)}$, the equilibrium constant for $A \rightleftharpoons P$ is:

(A) $K_{eq}^{(1)} / K_{eq}^{(2)}$ (B) $K_{eq}^{(2)} - K_{eq}^{(1)}$

(C) $K_{eq}^{(1)} + K_{eq}^{(2)}$ (D) $K_{eq}^{(1)} K_{eq}^{(2)}$

6. The gas phase reaction, $2NO_2(g) \longrightarrow N_2O_4(g)$ is an exothermic reaction. The decomposition of N_2O_4, in equilibrium mixture of $NO_2(g)$ and $N_2O_4(g)$, can be increased by:

(A) addition of an inert gas at constant pressure

(B) lowering the temperature

(C) increasing the pressure

(D) addition of an inert gas at constant volume

7. The value of K_c is 64 at 800 K for the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$. The value of K_c for the following reaction is:

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

(A) 1/64 (B) 8

(C) 1/4 (D) 1/8

8. For the reaction,

$$Fe_2N(s) + \frac{3}{2}H_2(g) \rightleftharpoons 2Fe(s) + NH_3(g)$$

(A) $K_c = K_p(RT)$

(B) $K_c = K_p(RT)^{\frac{-1}{2}}$

(C) $K_c = K_p(RT)^{\frac{1}{2}}$

(D) $K_c = K_p(RT)^{\frac{3}{2}}$

9. The variation of equilibrium constant with temperature is given below:

Temperature	Equilibrium Constant
$T_1 = 25\ ^oC$	$K_1 = 10$
$T_2 = 100\ ^oC$	$K_2 = 100$

The values of ΔH^o, ΔG^o at T_1 and ΔG^o at T_2 (in kJ mol^{-1}) respectively, are close to:

[Use R = 8.314 J K^{-1} mol^{-1}]

[JEE Main-2020]

(A) 28.4, –7.14 and –5.71

(B) 0.64, –7.14 and –5.71

(C) 28.4, –5.71 and –14.29

(D) 0.64, –5.71 and –14.29

10. For a reaction at equilibrium,

$$A(g) \rightleftharpoons B(g) + \frac{1}{2}C(g)$$

the relation between dissociation constant (K), degree of dissociation (α) and equilibrium pressure (p) is given by:

(A) $K = \dfrac{\alpha^{\frac{1}{2}} p^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$

(B) $K = \dfrac{\alpha^{\frac{3}{2}} p^{\frac{1}{2}}}{(2+\alpha)^{\frac{1}{2}}(1-\alpha)}$

(C) $K = \dfrac{(\alpha p)^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$

(D) $K = \dfrac{(\alpha p)^{\frac{3}{2}}}{(1+\alpha)(1-\alpha)^{\frac{1}{2}}}$

11. Ka_1, Ka_2 and Ka_3 are the respective ionization constants for the following reactions (a), (b) and (c).

(a) $H_2C_2O_4 \rightleftharpoons H^+ + HC_2O_4^-$

(b) $HC_2O_4^- \rightleftharpoons H^+ + HC_2O_4^{2-}$

(c) $H_2C_2O_4 \rightleftharpoons 2H^+ + C_2O_4^{2-}$

The relationship between K_{a_1}, K_{a_2} and K_{a_3} is given as:

(A) $K_{a_3} = K_{a_1} + K_{a_2}$ (B) $K_{a_3} = K_{a_1} - K_{a_2}$

(C) $K_{a_3} = K_{a_1} / K_{a_2}$ (D) $K_{a_3} = K_{a_1} \times K_{a_2}$

12. For the given reaction, choose the correct expression of K_C from the following:

$$Fe^{3+}_{(aq)} + SCN^-_{(aq)} \rightleftharpoons FeSCN^{2+}_{(aq)}$$

(A) $K_C = \dfrac{\left[FeSCN^{2+}\right]^2}{\left[Fe^{3+}\right]\left[SCN^-\right]}$

(B) $K_C = \dfrac{\left[FeSCN^{2+}\right]}{\left[Fe^{3+}\right]\left[SCN^-\right]}$

(C) $K_C = \dfrac{\left[FeSCN^{2+}\right]}{\left[Fe^{3+}\right]^2\left[SCN^-\right]^2}$

(D) $K_C = \dfrac{\left[Fe^{3+}\right]\left[SCN^-\right]}{\left[FeSCN^{2+}\right]}$

13. The correct relationship between K_P, α and equilibrium pressure P is:

$$A(g) \rightleftharpoons B(g) + \frac{C}{2}(g)$$

(A) $K_P = \dfrac{\alpha^{1/2}P^{3/2}}{(2+\alpha)^{3/2}}$

(B) $K_P = \dfrac{\alpha^{1/2}P^{1/2}}{(2+\alpha)^{1/2}}$

(C) $K_P = \dfrac{\alpha^{1/2}P^{1/2}}{(2+\alpha)^{3/2}}$

(D) $K_P = \dfrac{\alpha^{3/2}P^{1/2}}{(2+\alpha)^{1/2}(1-\alpha)}$

14. The following reaction occurs in the Blast Furnace where iron ore is reduced to iron metal:

$Fe_2O_3(s) + 3CO(g) \rightleftharpoons 2Fe(l) + 3CO_2(g)$

Using the Le Chatelier's principle, predict which one of the following will not disturb the equilibrium?

(A) Removal of CO

(B) Removal of CO_2

(C) Addition of CO_2

(D) Addition of Fe_2O_3

15. The equilibrium constant for the reaction

$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$ is $K_C = 4.9 \times 10^{-2}$.

The value of K_C for the reaction given below is

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is:

(A) 4.9 (B) 41.6

(C) 49 (D) 416

16. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Enthalpy of neutralisation of strong monobasic acid with strong monoacidic base is always –57 kJ mol^{-1}

Reason (R): Enthalpy of neutralisation is the amount of heat liberated when one mole of H^+ ions furnished by acid combine with one mole of OH^- ions furnished by base to form one mole of water.

In the light of the above statements, choose the correct answer from the options given below.

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true but (R) is not the correct explanation of (A)

17. The ratio $\frac{K_P}{K_C}$ for the reaction:

$CO_{(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons CO_{2(g)}$ is:

(A) $(RT)^{1/2}$ (B) RT

(C) 1 (D) $\frac{1}{\sqrt{RT}}$

18. At –20°C and 1 atm pressure, a cylinder is filled with equal number of H_2. I_2 and HI molecules for the reaction

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$,

the K_P for the process is $x \times 10^{-1}$. x = _____.

[Given: R = 0.082 L atm K^{-1} mol^{-1}]

(A) 2 (B) 1

(C) 10 (D) 0.01

19. For a reaction $A \xrightarrow{K_1} B \xrightarrow{K_2} C$

If the rate of formation of B is set to be zero then the concentration of B is given by:

(A) $K_1K_2[A]$ (B) $(K_1–K_2)[A]$

(C) $(K_1 + K_2)[A]$ (D) $(K_1/K_2)[A]$

20. $A_{(g)} \rightleftharpoons B_{(g)} + \frac{C}{2}(g)$

The correct relationship between K_P, α and equilibrium pressure P is

(A) $K_P = \frac{\alpha^{1/2}P^{1/2}}{(2+\alpha)^{1/2}}$

(B) $K_P = \frac{\alpha^{3/2}P^{1/2}}{(2+\alpha)^{1/2}(1-\alpha)}$

(C) $K_P = \frac{\alpha^{1/2}P^{3/2}}{(2+\alpha)^{3/2}}$

(D) $K_P = \frac{\alpha^{1/2}P^{1/2}}{(2+\alpha)^{3/2}}$

21. If 50mL of 0.5 M oxalic acid is required to neutralise 25 mL of NaOH solution, the amount of NaOH in 50 mL of given NaOH solution is g.

22. The following concentrations were observed at 500 K for the formation of NH_3 from N_2 and H_2. At equilibrium;

$[N_2] = 2\times10^{-2}M, [H_2] = 3\times10^{-2}M$ and $[NH_3] = 1.5\times10^{-2}M$. Equilibrium constant for the reaction is.

23. The following concentrations were observed at 500 K for the formation of NH_3 from N_2 and H_2. At equilibrium: $[N_2] = 2 \times 10^{-2}$ M, $[H_2] = 3 \times 10^{-2}$ M and $[NH_3] = 1.5 \times 10^{-2}$ M. Equilibrium constant for the reaction is _____.

24. For the reaction, $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, K_p = 0.492 atm at 300K. K_C for the reaction at same temperature is _____ $\times 10^{-2}$.

(Given: $R = 0.082 L atm mol^{-1} K^{-1}$)

25. NO_2 required for a reaction is produced by decomposition of N_2O_5 in CCl_4 as by equation $2N_2O_{5(g)} \rightarrow 4NO_{2(g)} + O_{2(g)}$

The initial concentration of N_2O_5 is $3 mol L^{-1}$ and it is $2.75 mol L^{-1}$ after 30 minutes.

The rate of formation of NO_2 is $x \times 10^{-3} mol L^{-1}$ min^{-1}, value of x is _____.

26. PCl_5 dissociates as
$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; 5 moles of PCl_5 are placed in a 200 litre vessel which contains 2 moles of N_2 and is maintained at 600 K. The equilibrium pressure is 2.46 atm. The equilibrium constant K_p for the dissociation of PCl_5 is _____ $\times 10^{-3}$ (nearest integer)
(Given: R = 0.082 L atm K^{-1} mol^{-1} : Assume ideal gas behaviour)

27. $2O_3(g) \rightleftharpoons 3O_2(g)$
At 300 K, ozone is fifty percent dissociated. The standard free energy change at this temperature and 1 atm pressure is (–) __J mol^{-1} (Nearest integer)
[Given: ln 1.35 = 0.3 and R = 8.3 J K^{-1} mol^{-1}]

28. The standard free energy change ($\Delta G°$) for 50% dissociation of N_2O_4 into NO_2 at 27°C and 1 atm pressure is –x J mol^{-1}. The value of x is _____. (Nearest Integer)
[Given: R = 8.31 J K^{-1} mol^{-1}, log 1.33 = 0.1239 ln 10 = 2.3]

29. 40% of HI undergoes decomposition to H_2 and I_2 at 300K. ΔG^- for this decomposition reaction at one atmosphere pressure is _____ J mol^{-1} (Nearest integer)
(Use R = 8.31 J K^{-1} mol^{-1}; log 2 = 0.3010.
ln 10 = 2.3, log 3 = 0.477)

30. The amount of charge in F (Faraday) required to obtain one mole of iron from Fe_3O_4 is _____. (Nearest Integer)

31. At 600K, 2 mol of NO are mixed with 1 mol of O_2. $2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)}$
The reaction occurring as above comes to equilibrium under a total pressure of 1 atom. Analysis of the system shows that 0.6 mol of oxygen are present at equilibrium. The equilibrium constant for the reaction is _____. (Nearest integer).

32. 0.01 moles of a weak acid HA ($K_a = 2.0 \times 10^{-6}$) is dissolved in 1.0 L of 0.1 M HCl solution. The degree of dissociation of HA is _____ $\times 10^{-5}$. (Round off to the Nearest Integer).
[Neglect volume change on adding HA. Assume degree of dissociation <<1]

33. At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture. The value of K_P for the reaction $Cl_{2(g)} \rightleftharpoons 2Cl_{(g)}$ under the above conditions is x $\times 10^{-1}$. The value of x is _____.
(Rounded of to the nearest integer)

34. For the reaction $A_{(g)} \rightarrow (B)_{(g)}$, the value of the equilibrium constant at 300 K and 1 atm is equal to 100.0. The value of $\Delta_r G$ for the reaction at 300 K and 1 atm in J mol^{-1} is –xR, where x is _____.
(Rounded of to the nearest integer)
(R = 8.31 J mol^{-1} K^{-1} and ln 10 = 2.3)

35. The stepwise formation of $\left[Cu(NH_3)_4\right]^{2+}$ is given below

$$Cu^{2+} + NH_3 \overset{K_1}{\rightleftharpoons} \left[Cu(NH_3)\right]^{2+}$$

$$\left[Cu(NH_3)\right]^{2+} + NH_3 \overset{K_2}{\rightleftharpoons} \left[Cu(NH_3)_2\right]^{2+}$$

$$\left[Cu(NH_3)_2\right]^{2+} + NH_3 \overset{K_3}{\rightleftharpoons} \left[Cu(NH_3)_3\right]^{2+}$$

$$\left[Cu(NH_3)_3\right]^{2+} + NH_3 \overset{K_4}{\rightleftharpoons} \left[Cu(NH_3)_4\right]^{2+}$$

The value of stability constants K_1, K_2, K_3 and K_4 are 10^4, 1.58×10^3, 5×10^2 and 10^2 respectively. The overall equilibrium constants for dissociation of $\left[Cu(NH_3)_4\right]^{2+}$ is x $\times 10^{-12}$. The value of x is _____. (Rounded off to the nearest integer)

36. Consider the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$. The temperature at which $K_C = 20.4$ and $K_P = 600.1$, is K.

(Round off to the Nearest Integer).

[Assume all gases are ideal and R = 0.0831 L bar $K^{-1} mol^{-1}$]

37. A homogeneous ideal gaseous reaction, $AB_{2(g)} \rightleftharpoons A_{(g)} + 2B_{(g)}$ is carried out in a 25 litre flask at 27°C. The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm. The value of K_P is $x \times 10^{-2}$. The value of x is_____. (Integer answer)

38. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$;

In an equilibrium mixture, the partial pressures are $P_{SO_3} = 43$ kPa; $P_{O_2} = 530$ Pa and $P_{SO_2} = 45$ kPa.

The equilibrium constant K_P = _____ $\times 10^{-2}$. (Nearest integer)

39. For the reaction, $A(g) \rightleftharpoons B(g)$ at 495 K, $\Delta_r G^0 = -9.478$ kJ mol^{-1}; If we start the reaction in a closed container at 495 K with 22 millimoles of A, the amount of B in the equilibrium mixture is _____ millimoles. (Round off to the Nearest Integer).

[R = 8.314 J mol^{-1} K^{-1}; $\ell n\ 10 = 2.303$]

40. Value of K_P for the equilibrium reaction, $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ at 288 K is 47.9. The K_C for this reaction at same temperature is _____. (Nearest integer)

(R = 0.083 L bar K^{-1} mol^{-1})

NSEJS PYQ

1. 0.1 mole of a weak acid HA and 0.1 mole of its sodium salt are dissolved in water and the solution is made upto 100 cm^3.

If $K_a = 1.8 \times 10^{-5}$, the pH of the solution is:

[NSEJS-2010]

(A) 2.4 (B) 3.70

(C) 4.74 (D) 5.00

2. Increase in pressure shifts the equilibrium of the reaction **[NSEJS-2010]**

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(A) to produce more N_2

(B) to produce more NH_3

(C) to reduce the temperature of the reaction

(D) to produce more H_2

3. The pKa of aspirin, a weak acid, is 3.5. The pH of gastric juice in the human stomach is between 2 and 3, while that in the small intestine is about 8. Aspirin will be...

[NSEJS-2011]

(A) unionized in the small intestine and the stomach

(B) completely ionized in the small intestine and the stomach

(C) ionized in the small intestine and almost unionized in the stomach

(D) ionized in the stomach and almost unionized in the small intestine

4. In the Haber process for the synthesis of ammonia, amount of ammonia formed will be more if: **[NSEJS-2011]**

(A) pressure is decreased and temperature is increased

(B) pressure is increased and temperature is decreased

(C) both pressure and temperature are increased

(D) both pressure and temperature are decreased

5. The equilibrium constant for the reaction **[NSEJS-2012]**

$N_2(g) + O_2(g) \Leftrightarrow 2NO(g)$

is 4.0×10^{-4} at 2000 K. In presence of a catalyst equilibrium is attained 10 times faster. Therefore, equilibrium constant in presence of catalyst at 2000 K is

(A) 40×10^{-4} (B) 4×10^{-4}

(C) 4×10^{-2} (D) none of these

6. The following reaction is at equilibrium in a given cylinder of constant volume. **[NSEJS-2012]**

$N_2(g) + 3H_2(g) \Leftrightarrow 2NH_3(g)$

Addition of argon gas to the cylinder will...

(A) reduce the formation of NH_3 with no change in temperature of the equilibrium system.

(B) Increase the formation of NH_3 with increase in temperature.

(C) reduce the formation of NH_3 with increase in temperature.

(D) increase the formation of NH_3 with no change in temperature of the equilibrium system.

7. A mixture of 0.3 moles of H_2 and 0.3 moles of I_2 are allowed to react in 10 litre evacuated flask at 500°C. The reaction is $H_2 + I_2 \Leftrightarrow 2HI$. Value of k is found to be 64. The amount of I_2 present at equilibrium is: **[NSEJS-2012]**

(A) 0.15 mole (B) 0.06 mole

(C) 0.03 mole (D) 0.2 mole

8. The equilibrium constant for the gaseous reaction $N_2 + O_2 \rightleftharpoons 2NO$ is k. The equilibrium constant for the formation of one mole of NO will be **[NSEJS-2012]**

(A) k/2 (B) k

(C) 2k (D) $\sqrt{k}$

9. The solubility of a salt B_2D_3 is X mol L^{-1}. Its solubility product is **[NSEJS-2012]**

(A) X^5 (B) $6X^5$

(C) $108X^5$ (D) $36X^5$

10. Which of the following will change the value of the equilibrium constant for the reaction?

$N_2(g) + O_2(g) \rightarrow 2NO(g)$ **[NSEJS-2014]**

(A) Add more N_2

(B) Increase of pressure

(C) Use a smaller reaction vessel

(D) Increase the temperature

11. For the reaction $aE + bF \rightleftharpoons cG + dH$.

The expression of equilibrium constant can be written as **[NSEJS-2014]**

$$K_c = [G]^c [H]^d / [E]^a [F]^b$$

Where all the concentrations are equilibrium concentrations. Before approaching equilibrium, the same concentration ratio is called reaction quotient Q_c.

For the reaction system to reach equilibrium

(A) Q_c must increase in the reaction

(B) Q_c must decrease in the reaction

(C) $Q_c = K_c$

(D) Q_c = zero

12. In Haber's process 0.240 mole of Nitrogen, 3.9 moles of hydrogen are taken which lead to the formation of 7.8 moles product in a 3.00 litres of reaction vessel at 375°C. Considering that equilibrium constant at this temperature is 41.2 Calculate the value of reaction quotient (Q) and predict whether the reaction is in equilibrium or it will proceed in either direction. **[NSEJS-2014]**

(A) Q = 38.62 and reaction will be in equilibrium.

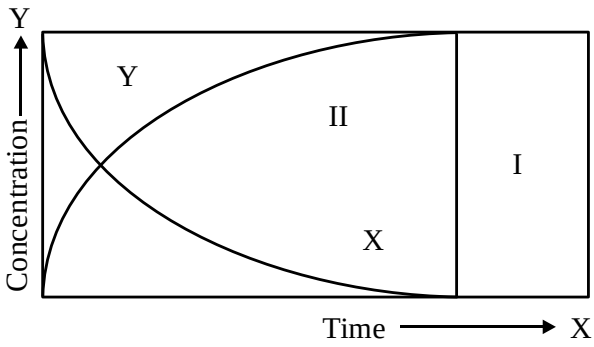
(B) Q = 19.31 and reaction will proceed in forward direction.

(C) Q = 38.62 and reaction will proceed in forward direction.

(D) Q = 19.31 and reaction will proceed in backward direction.

13. Variation of the concentration of the reactant (X) and the product (Y) are shown in the figure. Select the correct statement.

[NSEJS-2014]



(A) I and II both are kinetic regions.
(B) I and II both are equilibrium regions.
(C) I is equilibrium and II is kinetic region.
(D) I is kinetic and II is equilibrium region.

14. At 5 atm pressure PCl_5 gas dissociates by 10%. What will be the value of K_p at same temperature? **[NSEJS-2014]**

(A) 0.045 atm (B) 0.050 atm
(C) 0.9 atm (D) 0.5 atm

15. **Assertion (A):** If the volume of the vessel is doubled then for the following reaction.

$A(g) \rightarrow B(g) + C(g)$ **[NSEJS-2014]**

Equilibrium constant is decreased.

Reason (R): Equilibrium constant

$K_c = X^2/(1 - X)V$

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)
(B) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(C) (A) is true but (R) is false
(D) (A) is false but (R) is true

16. At 5 atm pressure PCl_5 gas dissociates by 10%. What will be the value of K_p at same temperature? **[NSEJS-2014]**

(A) 0.045 atm (B) 0.050 atm
(C) 0.9 atm (D) 0.5 atm

17. Consider the following reaction:

$4\ PCl_3(g) \longrightarrow P_4(g) + 6Cl_2(g)$. If the initial concentration of $PCl_3(g)$ is 1.0 M, and "x" is the equilibrium concentration of $P_4(g)$, what is the correct equilibrium relation?

[NSEJS-2014]

(A) $K_c = 6x^7$
(B) $K_c = 6x^7 / (1.0 - x)^4$
(C) $K_c = (x)(6x)^6 / (1.0 - 4x)^4$
(D) $K_c = x^7 (1.0 - x)^4$

18. A bottle of 200 ml of hydrogen peroxide was accidentally left open for long time. Which of the following facts hold good for the chemical reaction that takes place?

[NSEJS-2015]

(K_p is an equilibrium constant in terms of partial pressures of reactants and products and Kc is equilibrium constant in terms of partial pressures of reactants and products and K_c is equilibrium constant in terms of molar concentrations of reactants and products).

(A) $K_p - K_c$
(B) $K_p > K_c$
(C) $K_p < K_c$
(D) K_p and K_c cannot be correlated

19. During a meteorite shower a few meteorites fell into a water body having pH around 7. The pH of the water body was measured after meteorite shower and found to be

[NSEJS-2018]

(A) > 7
(B) < 7
(C) = 7
(D) no change in pH of water due to the meteorite shower

20. Arrange the following solutions in increasing hydronium ion concentration. The solutions are: **[NSEJS-2020]**

(P) 0.1 M HCl
(Q) 0.1 M H_2SO_4
(R) 0.001 M NH_4OH
(S) 0.001 M $Ca(OH)_2$

The correct order will be:

(A) P > Q > R > S (B) Q > P > S > R
(C) S > R > Q > P (D) S > R > P > Q

Advanced & Olympiad Challenger Questions

1. An aqueous buffer is prepared by adding 100 ml of 0.1 mol 1^{-1} acetic acid to 50 ml of 0.2 mol l^{-1} of sodium acetate. If pKa of acetic acid is 4.76, the pH of the buffer is

(A) 4.26 (B) 5.76

(C) 3.76 (D) 4.76

2. At 783 K in the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, the molar concentrations (mol 1) of H_2, I_2 and HI at some instant of time are 0.1, 0.2 and 0.4, respectively. If the equilibrium constant is 46 at the same temperature, then as the reaction proceeds

(A) the amount of HI will increase

(B) the amount of HI will decrease

(C) the amount of H_2 and I_2 will increase

(D) the amount of H_2 and I_2 will not change

3. A reaction with reaction quotient Q_C and equilibrium constant K_C, will proceed in the direction of the products when:

(A) $Q_C = K_C$ (B) $Q_C < K_C$

(C) $Q_C > K_C$ (D) $Q_C = 0$

4. The equilibrium constant, K_C for is 4 L^2 mol^{-2}. If the equilibrium concentration of benzene is 0.5 mol L^{-1}, that of acetylene in mol L^{-1}must be:

(A) 0.025 (B) 0.25

(C) 0.05 (D) 0.5

5. The degree of dissociation of acetic acid (0.1 mol L^{-1}) in water (K_a of acetic acid is 10^{-5}) is:

(A) 0.01

(B) 0.5

(C) 0.1

(D) 1.0

6. The solubility of $BaSO_4$ in puree water (in g L^{-1}) is closest to-

[Given: K_{sp} of $BaSO_4$ is 1.0×10^{-10} at 25°C. Molecular weight of $BaSO_4$ is 233 g mol^{-1}]

(A) 1.0×10^{-5}

(B) 1.0×10^{-3}

(C) 2.3×10^{-5}

(D) 2.3×10^{-3}

7. Read the statements given below:

(i) Dissolution of glucose in water is an exothermic process.

(ii) Mixing of calcium oxide in water is an endothermic process

(iii) Melting of ice into water is an endothermic process.

(iv) Dilution of sluphuric acid in water is an endothermic process.

(v) Boiling of water is an exothermic process.

Of the above, the true statement/s is/are:

(A) (iii)

(B) (v) and (i)

(C) (iv) and (v)

(D) (i), (ii), and (iv)

Answer Key

EXERCISE-1

1. (D) **2.** (C) **3.** (B) **4.** (D)
5. (B) **6.** (C) **7.** (D) **8.** (C)
9. (D) **10.** (C) **11.** (C) **12.** (D)
13. (D) **14.** (A) **15.** (C) **16.** (C)
17. (A) **18.** (B) **19.** (B) **20.** (C)
21. (C) **22.** (D) **23.** (A) **24.** (D)
25. (B) **26.** (D) **27.** (C) **28.** (D)
29. (D) **30.** (C) **31.** (C) **32.** (A)
33. (D) **34.** (A) **35.** (A) **36.** (B)
37. (A) **38.** (D) **39.** (B) **40.** (A)
41. (D) **42.** (D) **43.** (D) **44.** (A)
45. (B) **46.** (A) **47.** (D) **48.** (C)
49. (B) **50.** (B) **51.** (C) **52.** (B)
53. (B) **54.** (B) **55.** (C) **56.** (D)
57. (B) **58.** (A) **59.** (C) **60.** (A)
61. (A) **62.** (C)

EXERCISE-2

1. (B) **2.** (C) **3.** (A) **4.** (D)
5. (D) **6.** (A) **7.** (D) **8.** (C)
9. (C) **10.** (B) **11.** (D) **12.** (B)
13. (D) **14.** (D) **15.** (D) **16.** (B)
17. (D) **18.** (C) **19.** (D) **20.** (B)
21. (4) **22.** (4) **23.** (417) **24.** (2)
25. (17) **26.** (1107) **27.** (747) **28.** (710)
29. (2735) **30.** (3) **31.** (2) **32.** (2)
33. (5) **34.** (1380) **35.** (1) **36.** (354)
37. (74 or 72 to 75) **38.** (17228) **39.** (20) **40.** (2)

NSEJS PYQ

1. (C) **2.** (B) **3.** (C) **4.** (B)
5. (B) **6.** (C) **7.** (B) **8.** (D)
9. (C) **10.** (D) **11.** (C) **12.** (C)
13. (C) **14.** (D) **15.** (A) **16.** (D)
17. (C) **18.** (B) **19.** (A)
20. (**No option is correct and it should be** Q > P > R > S)

Advanced & Olympiad Challenger Questions

1. (D) **2.** (A) **3.** (B) **4.** (D)
5. (D) **6.** (D) **7.** (D)